BDA 411/511 Introduction to Machine Learning: Midterm Review Notes

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1. Linear algebra, SVD and PCA

Symmetric positive semi-definite (PSD) matrix. A symmetric A is PSD if $\mathbf{z}^T A\mathbf{z} \geq 0$ for all $\mathbf{z}$, equivalently all eigenvalues are ≥ 0. It is positive definite (PD) if all eigenvalues are > 0 (Sylvester's test: all leading principal minors are positive).

- A PSD matrix does *not* need nonnegative entries. Example: $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ has eigenvalues 0 and 2.
- The eigenvectors of a symmetric matrix can be chosen orthonormal. For a non-symmetric matrix they generally are not orthogonal.

SVD. Any $X \in \mathbb{R}^{n \times d}$ factors as $X = U\Sigma V^T$ with orthonormal columns in U and V and singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.

PCA. Center the data first. The principal directions are the right singular vectors V of the centered data matrix. They are also the eigenvectors of the covariance matrix $C = \frac{1}{n-1} X^T X$, and $\lambda_j = \sigma_j^2/(n-1)$. So PCA can be computed by *either* the SVD of the data matrix *or* the eigendecomposition of the covariance matrix.

Truncated SVD. $X_r = \sum_{j \leq r} \sigma_j \mathbf{u}_j \mathbf{v}_j^T$ is the best rank-r approximation. Its squared Frobenius error is $\sum_{j>r} \sigma_j^2$. The fraction of energy kept is $\sum_{j \leq r} \sigma_j^2 / \sum_j \sigma_j^2$.

2. Probability tools

- **Law of total probability:** $P(B) = \sum_i P(B \mid A_i) P(A_i)$.
- **Bayes' rule:** $P(A \mid B) = \dfrac{P(B \mid A) P(A)}{P(B)}$. Posterior $\propto$ likelihood $\times$ prior.
- For independent, identically distributed (iid) data, the likelihood factorizes: $p(\mathbf{y} \mid X; \theta) = \prod_{i=1}^n p(y_i \mid x_i; \theta)$.
- Useful density: the exponential distribution $p(y; \theta) = \theta e^{-\theta y}$ for $y \geq 0$.

3. Linear regression

Model. With $x_i = (1, x_{i1}, \ldots, x_{id})^T$ (the constant 1 carries the intercept θ_0), the prediction is $\hat{y}_i = \theta^T x_i$.

Least-squares loss. $J(\theta) = \frac{1}{2} \sum_{i=1}^n (\theta^T x_i - y_i)^2 = \frac{1}{2} \|X\theta - \mathbf{y}\|_2^2$. A factor $\frac{1}{n}$ or $\frac{1}{2}$ in front is only a convention. It does not change the minimizer.

Gradient. $\nabla J(\theta) = \sum_i (\theta^T x_i - y_i) x_i = X^T(X\theta - \mathbf{y})$.

Update rules (learning rate $\alpha > 0$):

- *Batch gradient descent (BGD):* $\theta \leftarrow \theta - \alpha\, X^T(X\theta - \mathbf{y})$. Uses all n samples per step.
- *Stochastic gradient descent (SGD):* pick one sample i at random and set $\theta \leftarrow \theta - \alpha\,(\theta^T x_i - y_i)\, x_i$. Cheap and noisy. Mini-batch SGD sits in between.

Normal equation. Setting the gradient to zero gives $X^T X\theta = X^T\mathbf{y}$, so $\theta = (X^T X)^{-1} X^T \mathbf{y}$ when X has full column rank. To derive it, expand $J = \frac{1}{2}(\theta^T X^T X\theta - 2\theta^T X^T\mathbf{y} + \mathbf{y}^T\mathbf{y})$, take the gradient, and set it to 0.

Practical points. Too large a learning rate diverges. Too small is slow. Scaling the features improves conditioning. The loss is convex, so gradient descent reaches the global minimum when α is small enough.

4. The probabilistic view: MLE and MAP

Least squares is an MLE. If $y_i = \theta^T x_i + \epsilon_i$ with iid $\epsilon_i \sim N(0, \sigma^2)$, maximizing the likelihood is the same as minimizing $\sum_i (y_i - \theta^T x_i)^2$. Linear regression finds this MLE exactly when the mean function is *linear in the parameters* (it may be nonlinear in x, for example $\theta_1 \sin x$ or $\theta_1 x + \theta_2 x^2$).

Recipe for an MLE.

1. Write the likelihood $L(\theta) = \prod_i p(y_i \mid x_i; \theta)$.
2. Take the log: $\ell(\theta) = \sum_i \log p(y_i \mid x_i; \theta)$.
3. Differentiate, set $\ell'(\theta) = 0$, and solve.
4. Check the second derivative is negative so that it is a maximum.

Worked example (Bernoulli). For coin flips with k heads in n tosses, $\ell(p) = k \log p + (n - k)\log(1 - p)$. Then $\ell'(p) = \frac{k}{p} - \frac{n-k}{1-p} = 0$ gives $\hat{p}_{\text{MLE}} = k/n$.

MAP. The maximum a posteriori estimate maximizes $\log p(\mathbf{y} \mid X; \theta) + \log p(\theta)$, that is, log-likelihood plus log-prior. The key difference from MLE: MAP uses a prior and MLE does not. As the amount of data grows, the prior matters less and MAP approaches the MLE.

Regularization as a prior.

- Gaussian prior on the weights $\Longleftrightarrow$ L_2 (ridge) regularization: $\min_\theta \ \frac{1}{2}\|X\theta - \mathbf{y}\|^2 + \frac{\lambda}{2}\|\theta\|_2^2$, with solution $\theta = (X^T X + \lambda I)^{-1} X^T \mathbf{y}$.
- Laplace prior $\Longleftrightarrow$ L_1 (lasso) regularization, which encourages sparse weights.

5. Overfitting, underfitting and model selection

- **Underfitting** (high bias): the model is too simple. Both training and test error are large.

- **Overfitting** (high variance): the model is too flexible. Training error is small but test error is large.
- Using more features or a higher-degree polynomial *reduces* training error but does not guarantee better generalization.
- **Ways to reduce overfitting:** get more training data, add regularization, use a simpler model, use early stopping, and choose hyperparameters by validation or cross-validation.
- Split data into training, validation and test sets. Never tune on the test set.
- When comparing fitted curves, prefer the model that captures the overall trend without chasing every noisy point.

6. Logistic regression and decision boundaries

Model (labels $y \in \{0, 1\}$). $P(y = 1 \mid x) = \sigma(\theta^T x) = \dfrac{1}{1 + e^{-\theta^T x}}$.

Loss. The negative log-likelihood is the cross-entropy loss $J(\theta) = -\sum_i \left[y_i \log \hat{p}_i + (1-y_i) \log(1-\hat{p}_i)\right]$ with $\hat{p}_i = \sigma(\theta^T x_i)$. Its gradient is $X^T(\hat{\mathbf{p}} - \mathbf{y})$. There is no closed-form solution, so use gradient descent or a similar method.

Decision boundary. Predict class 1 when $\theta^T x > 0$. The boundary $\{x : \theta^T x = 0\}$ is a line in 2D and a hyperplane in general. It separates the classes in feature space. Logistic regression is a *linear* classifier, so on data that are not linearly separable it makes some errors unless you add nonlinear features or use a kernel.

7. Support vector machines (SVM)

Hard margin. For labels $y_i \in \{-1, +1\}$: $\min_{\mathbf{w},b} \ \frac{1}{2}\|\mathbf{w}\|^2$ subject to $y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1$. The margin (the total width of the empty slab) is $2/\|\mathbf{w}\|$. The points that lie exactly on the margin are the *support vectors*. They determine the solution.

No solution without separability. If the data are not linearly separable (in the chosen feature space), the hard-margin problem has no feasible solution.

Soft margin. Introduce slack variables $\xi_i \geq 0$: $\min \ \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_i \xi_i$ subject to $y_i(\mathbf{w}^T\mathbf{x}_i + b) \geq 1 - \xi_i$. Here $\xi_i = 0$ means the point respects the margin, $0 < \xi_i \leq 1$ means it is inside the margin but correctly classified, and $\xi_i > 1$ means it is misclassified. Large C penalizes violations strongly (less regularization). Small C allows a wider margin with more violations.

Hinge loss. $\ell_{\text{hinge}} = \max\big(0,\, 1 - y_i(\mathbf{w}^T\mathbf{x}_i + b)\big)$ is a convex upper bound on the 0–1 loss, and the soft-margin objective equals hinge loss plus $\frac{1}{2}\|\mathbf{w}\|^2$.

SVM versus logistic regression. Both give a linear boundary. SVM uses the hinge loss (zero loss for points beyond the margin, so only support vectors matter). Logistic regression uses the log loss (every point contributes) and gives probabilities.

8. Kernel methods

- **Why kernels:** they let linear methods model nonlinear relationships by working in a high-

dimensional feature space $\phi(x)$ without computing ϕ explicitly. Only the kernel $k(x, x') = \phi(x)^T\phi(x')$ is needed.

- **Common kernels:** polynomial $(x^Tx' + c)^p$ and radial basis function (RBF, Gaussian) $\exp\big(-\|x - x'\|^2/(2\sigma^2)\big)$.
- A valid kernel gives a symmetric positive semi-definite kernel (Gram) matrix.
- **Which methods can be kernelized:** SVM, linear regression (kernel ridge regression) and logistic regression (kernel logistic regression), among others.
- In the RBF kernel a small σ gives a wiggly, overfit boundary. A large σ gives an overly smooth, underfit one.

9. Quick-check list of facts

- PSD does not imply nonnegative entries.
- PCA can be computed by SVD of the data *or* by eigendecomposition of the covariance.
- Hard-margin SVM has no solution on non-separable data.
- Logistic regression is a classification method despite its name.
- More features do not always improve generalization.
- Underfitting means the model is too simple, not too complicated.
- MAP uses a prior. MLE does not.
- Slack variables measure margin violations (a point is misclassified when $\xi_i > 1$).
- Gaussian prior $\leftrightarrow$ L_2 regularization. Laplace prior $\leftrightarrow$ L_1.
- Overfitting: small training error, large test error (high variance, low bias).

10. Self-test questions

1. Show that $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ is symmetric positive definite in two different ways.
2. Derive the normal equation and the BGD and SGD update rules for $J(\theta) = \frac{1}{2}\|X\theta - \mathbf{y}\|^2$.
3. Derive the ridge-regression solution $\theta = (X^TX + \lambda I)^{-1}X^T\mathbf{y}$.
4. Observations $y_1, \ldots, y_n$ are iid Poisson with mean λ. Find the MLE of λ.
5. Observations $y_1, \ldots, y_n \sim N(\mu, 1)$ with prior $\mu \sim N(0, \tau^2)$. Find the MAP estimate of μ and describe what happens as $\tau \to \infty$.
6. A model has training error 1% and test error 25%. Name two ways to improve it.
7. What does the slack variable ξ_i mean in a soft-margin SVM? What does the parameter C control?

8. On a two-class plot with a straight decision boundary, how do you count misclassified points, and what does a straight boundary tell you about the method used?